31. Determine all x for which $f(x) = \frac{x+2}{x-1}$ is positive. Also find its range.

32. Let $f(x) = |x - 1| + |x + 1|$. Find its minimum value and the corresponding x.

33. Show that $f(x) = x^2$ is one-one only on [0, ∞) or (−∞, 0]. Justify.

34. For $f(x) = e^{-x^2}$, find domain, range, and maximum value.

35. Let R be a relation on integers defined by $aRb \iff 5|(a-b)$. Prove R is an equivalence relation.

36. Determine whether $f(x) = x^3 - 3x$ is one-one or onto. Find its inverse if possible.

37. If $f: \mathrm{R} \to \mathrm{R}$, $f(x) = x^2 + x + 1$, find the range using calculus.

38. Show that a function f(x) = x^5 + x + 1 is invertible on ℝ.

39. Find domain and range of $f(x) = \sqrt{1 - x^2} + \sqrt{2 + x}$.

40. Let $f(x) = \tan^{-1}(x)$. Show that f is one-one, onto (−π/2, π/2), and invertible.

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41. Determine the number of reflexive relations on a set with n elements.

42. Show that the function $f(x) = x/(1 + x)$ is strictly increasing and invertible.

43. Determine the composition $(g \circ f)(x)$ for $f(x) = 2x + 1$ and $g(x) = x^2$. Discuss domain and range.

44. Let $f(x) = \sqrt{x^2 + 1} - x$. Show that its range is (0,1].

45. Prove that the inverse of an odd, invertible function is also odd.

46. Consider $f(x) = |x|$. Find an interval on which f is invertible and compute f^{-1}.

47. Let R be “has same remainder when divided by 3” on integers. Show R is equivalence and determine equivalence classes.

48. For $f(x) = \ln(1 + e^x)$, find its domain and range.

49. Show that the function $f(x) = x^2 + 2x + 5$ attains its minimum at x = −1 and find the minimum.

50. If f(x) = x^2, g(x) = √x, find g∘f and f∘g, and discuss their domains.

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51. Let f be a bijection from set A to B. Prove that $f^{-1} \circ f(x) = x$ ∀x∈A and $f \circ f^{-1}(y) = y$ ∀y∈B.

52. Show that the sum of two one-one functions need not be one-one. Give an example.

53. For $f(x) = 1/(x - 2)$, find domain, range, and inverse.

54. Determine the range of $f(x) = x + 1/x$ for x>0 using calculus.

55. Let R be the relation "is perpendicular to" on lines in a plane. Is R symmetric? Reflexive? Transitive?

56. Prove that the composite of two bijective functions is bijective.

57. Let f(x) = |x-3| + |x+3|. Find its minimum value and the x at which it occurs.

58. Show that the function f(x) = x^2 + |x| has range [0, ∞).

59. Determine the domain of $\ln(\tan^{-1}(x))$ and sketch the function.

60. Prove that the function f(x) = e^x is invertible and find f^{-1}.

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61. For the relation R on ℝ defined by $aRb \iff ab \geq 0$, check reflexive, symmetric, transitive properties.

62. Determine the number of symmetric relations on a set with 3 elements.

63. If f(x) = cos(sin x), find exact range.

64. Show that f(x) = x^3 - x is not one-one on ℝ but is one-one on [0, ∞).

65. For the function $f(x) = x^2 - 4x + 7$, find the range by completing the square.

66. Determine all x for which $f(x) = \sqrt{x-1} + \sqrt{3-x}$ is defined.

67. Prove that the number of onto functions from set A (m elements) to B (n elements) is n! × S(m,n), where S(m,n) is Stirling number of second kind.

68. Let $f(x) = \sin^{-1}(x^2-1)$. Determine its domain and range.

69. Show that the function $f(x) = \ln(1+x^2)$ is strictly increasing and find its range.

70. Determine all x for which $f(x) = |x-1| + |x+2|$ is differentiable.

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71. For $f(x) = x/(1-x)$, find the inverse function and verify $f^{-1}(f(x)) = x$.

72. Prove that the identity function on any set is bijective.

73. For $f(x) = \sqrt{(x^2+1)} - x$, show $f(f(x)) = 1/(2x)$ and discuss domain.

74. Show that $f(x) = x^3 + x$ is invertible and find $f^{-1}(x)$ using approximate methods.

75. Determine domain and range of $f(x) = \ln(1-x^2)$.

76. If $f: \mathbb{R} \to \mathbb{R}$ is one-one, show that $f(x) = f(y) \Rightarrow x = y$.

77. Let R be the relation "divides" on natural numbers. Show it is reflexive and transitive but not symmetric.

78. Determine the range of $f(x) = x^2/(x^2+1)$ for all real x.

79. For $f(x) = \tan^{-1}(1/x)$, find domain, range, and derivative.

80. Prove that if f and g are one-one, then $g \circ f$ is one-one. Provide counterexample if one is not one-one.

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81. Let $f(x) = |x|$. Show that restricting f to $[0, \infty)$ makes it invertible and find f^{-1}.

82. Prove that a function $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 + x + 1$ is not onto.

83. Find domain and range of $f(x) = \sqrt{(x^2 - 2x - 3)}$.

84. Determine whether $f(x) = \cos^{-1}(x^2)$ is invertible. If not, restrict its domain to make it invertible.

85. Let R be "a is sibling of b" on a set of people. Discuss whether R is equivalence, symmetric, reflexive, transitive.

86. Find domain and range of $f(x) = x^2 + |x-1|$.

87. Show that the composite of two invertible functions is invertible, and find inverse in terms of f^{-1}, g^{-1}.

88. Determine the number of reflexive and symmetric relations on a 4-element set.

89. Prove that the function $f(x) = x^3 + 3x + 1$ is strictly increasing and invertible.

90. For $f(x) = |x| + |x-1| + |x-2|$, find domain, range, and minimum value.

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91. Determine whether $f(x) = x^3 - x + 1$ is one-one. If so, find f^{-1}.

92. Show that the function $f(x) = \sqrt{(x^2+4x+5)}$ has minimum 1 and find x where it occurs.

93. Find all x such that $f(x) = \ln(1/(x-1))$ is defined.

94. Determine domain and range of $f(x) = x + \sqrt{(x^2+1)}$.

95. Prove that the inverse of a monotone increasing function is also monotone increasing.

96. Determine the range of $f(x) = 1/(1 - \sin x)$ for all x in its domain.

97. For $f(x) = \sqrt{(2x-3)}$, find domain and range, and discuss monotonicity.

98. Let R be the relation "has the same parity" on $\mathbb{Z}$. Show it is an equivalence relation.

99. Show that the function $f(x) = x^2 + 2x + 2$ attains its minimum at $x = -1$.

100. Prove that the set of all bijections from $A \to A$ has cardinality $n!$ if $|A|=n$.

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101. For $f(x) = \cos(\sin x)$, find exact domain and range.

102. Determine all x such that $f(x) = \sqrt{x-\sqrt{x}}$ is defined.

103. Show that $f(x) = x^3 - 3x$ is invertible on $[\sqrt{3}, \infty)$.

104. Determine domain and range of $f(x) = x^2 - 2|x| + 3$.

105. Prove that the sum of two one-one functions need not be one-one.

106. Determine the domain and range of $f(x) = \ln(1+x^2)$.

107. Show that the function $f(x) = |x-1| + |x+2|$ is convex.

108. Prove that the relation $R = \{(x,y): x^2 + y^2 = 1\}$ is symmetric but not transitive.

109. Show that the function $f(x) = x^2/(x^2+1)$ is strictly increasing on $[0, \infty)$ and decreasing on $(-\infty,0)$.

110. Determine the domain of $f(x) = \ln(\ln x)$.

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111. For $f(x) = |x| + |x-2|$, determine intervals where f is differentiable.

112. Show that $f(x) = x^3 - 3x + 1$ is strictly increasing and find inverse approximately.

113. Let $f(x) = x/(1+x)$. Show that it is bijective and find f^{-1}.

114. Determine the domain and range of $f(x) = \sqrt{x^2 - 5x + 6}$.

115. For $f(x) = \cos^{-1}(x^2)$, find domain and range.

116. Show that the function $f(x) = \ln(1 + e^x)$ is one-one and onto $(0, \infty)$.

117. Determine the domain of $f(x) = \sqrt{4 - |2x - 1|}$.

118. Prove that $f(x) = x^2 + 2x + 5$ has minimum value using calculus.

119. Let $f(x) = x^2 + |x|$. Show that its range is $[0, \infty)$.

120. For $f(x) = \tan^{-1}(1/x)$, determine domain, range, and inverse function.